

Implementing Belief-Function Networks for Fault Diagnosis: From Bayesian Network Baseline to Industrial Applicability

Pablo Garcia-Legaz Levi - 2727290

May 14, 2026

Contents

1	Introduction	3
2	Background and Related Work	5
2.1	Fault diagnosis under uncertainty	5
2.2	Bayesian probability and Bayesian networks	6
2.3	Belief functions and Dempster–Shafer theory	7
2.4	Belief-function graphical models	8
2.5	State of the art and research gap	9
3	Problem Statement and Research Scope	9
4	Case Study and Dataset	10
4.1	Electrical System Description	10
4.2	Fault and Observable Variables	11
4.3	Dataset Construction and Assumptions	11
4.4	Diagnostic Task	12
4.5	Bayesian Network Baseline	12
5	Belief-Function Implementation	12
5.1	From Bayesian Network to Belief-Function Model	13
5.2	Mass Functions, Evidence, and Inference	13
5.3	Implementation Using Belief Function Machine	13
5.4	Worked Example	13
6	Results and Analysis	13
6.1	Experimental Setup	13
6.2	Diagnostic Results	13
6.3	Comparison with the Bayesian Network Baseline	13
6.4	Interpretation of the Results	13

7	Applicability and Scalability	13
7.1	Practical Usefulness and Limitations	13
7.2	Scalability and Bottlenecks	14
7.3	Software and Tooling	14
8	Future Work???	14
9	Conclusion	14
A	Additional Dataset Details	14
B	Additional Bayesian Network Details	14
C	Additional Belief-Function Model Details	14
D	Additional Results	14

1 Introduction

Modern **technical systems** are increasingly complex, interconnected, and dependent on reliable operation. In domains such as industrial machinery, electronic systems, transportation, energy systems, and cyber-physical systems, a failure in one component can propagate through the system and lead to degraded performance, safety risks, financial losses, or complete system shutdown. When such a system fails, it is not sufficient to observe that the system is not working; engineers must identify the underlying cause of the observed symptoms. This task is known as fault diagnosis. In practical terms, fault diagnosis aims to infer which component, connection, or subsystem is responsible for a set of observed abnormal behaviours.

Fault diagnosis is especially important in **enterprise** contexts, where service engineers often need to make decisions under time pressure and with incomplete information. A diagnosis application must combine available information about the system structure, observed symptoms, sensor readings, and expert knowledge in order to suggest likely root causes. However, real diagnostic situations rarely provide perfect information. Observations may be noisy, components may be hidden, data may be scarce, and expert knowledge may be qualitative rather than numerical. These characteristics make fault diagnosis a problem of reasoning under uncertainty.

 Bayesian networks are one of the most widely used probabilistic graphical models for **reasoning under uncertainty**. A Bayesian network represents variables as nodes in a directed acyclic graph, where edges encode probabilistic dependencies between variables. Each variable is associated with a prior probability or a conditional probability table, and inference is performed by updating probabilities after evidence is observed. This makes Bayesian networks attractive for fault diagnosis because they provide a structured way to represent causal or diagnostic relationships between faults and symptoms. **In the context of this project**, a Bayesian network is used as the **collaborative** baseline model for a small electrical fault-diagnosis system consisting of a power supply module, eight control modules, and a load module.

Despite their usefulness, Bayesian networks also impose an important requirement: the relevant probability distributions must be specified precisely. In a complete Bayesian network, every prior and every conditional probability table must be available. **This requirement can become problematic in real enterprise diagnosis problems**. For example, a company may know that a switch failure can prevent a lamp from turning on, but it may not know the exact conditional probability of each possible symptom under each possible fault. Similarly, engineers may know that a missing indicator light provides evidence about a local electrical interruption, but they may not be able to quantify this evidence with a single precise probability. **If these unknown probabilities are replaced by arbitrary assumptions, such as uniform distributions or maximum-entropy estimates, the resulting model may produce precise-looking answers that hide the underlying uncertainty.**

Dempster-Shafer theory, also known as belief function **theory**, offers an alternative framework for representing uncertainty. Instead of assigning probability mass only to single hypotheses, belief function theory **allows mass to be assigned to sets of hypotheses**. This makes it possible to represent partial ignorance explicitly. From a mass function, two quantities can be derived: belief, which measures the support that is definitely committed to a proposition,

and plausibility, which measures the support that remains possible for that proposition. As a result, conclusions can be expressed as intervals rather than as single probabilities. In a diagnostic setting, this means that the model can express not only which fault is supported by the evidence, but also how much uncertainty remains.

This thesis focuses on the use of belief-function networks for fault diagnosis, based especially on Shenoy’s Dempster-Shafer approach to inference in incomplete Bayesian networks. Shenoy proposes embedding an incomplete Bayesian network into a corresponding belief-function model, omitting unavailable probability information rather than replacing it with artificial point estimates. In this way, the Bayesian network structure can still be used as a graphical representation of dependencies, while belief functions allow incomplete priors or conditionals to be represented as ignorance. The resulting inference produces probability intervals, which can be more informative than a single posterior probability when the model itself is incomplete.

[Not Sure if this would go here or later in the description of the dataset] The case study used in this thesis is a small electrical system composed of ten modules connected in series: one power supply module, eight control modules, and one load module. The power supply contains a battery and a power indicator. Each control module contains a switch and a switch indicator LED. The load module contains the lamp whose operation is the main system output. The diagnostic problem is to infer which component or connection may be faulty given observations such as whether the power indicator, switch indicators, and load lamp are on or off. This system is intentionally simple, but it captures an important feature of real diagnostic systems: Observable symptoms are only indirect evidence of the underlying fault state.

The aim of this thesis is not only to apply belief function theory mathematically but also to evaluate its practical usefulness as a diagnostic approach and readiness at an industry level. The intended contribution is therefore methodological and explanatory. First, the thesis shows how a Bayesian-network representation of the electrical fault-diagnosis system can be used as a basis for a belief-function model. Second, it explains how incomplete probabilistic information can be handled using Dempster-Shafer belief functions and implemented using the Belief Function Machine in MATLAB [For now the best and most complete approach literature review still ongoing]. Third, it analyzes whether this approach is suitable for enterprise use considering its advantages, limitations, and level of technology readiness.

This work differs from existing applications of Dempster-Shafer theory to fault diagnosis in three main ways. Many existing studies apply Dempster-Shafer theory mainly as an evidence fusion technique, for example by combining sensor outputs or classifier predictions. Other works focus on improving diagnostic accuracy in a specific technical domain. By contrast, this thesis focuses on the connection between Bayesian networks and belief-function networks for incomplete diagnostic models. It also aims to be understandable for a potential practitioner: the goal is to explain when the approach is useful, how it can be implemented, and what limitations an enterprise should consider before adopting it. Finally, the thesis includes a technology-readiness perspective, asking not only whether the method works in a toy system, but how mature and transferable it is as a practical diagnostic technology.

The research question guiding this thesis is:

How can a Bayesian Network fault-diagnosis model be transformed into a Dempster-

Shafer belief-function network, and what does this transformation reveal about the practical applicability, scalability, and technology readiness of belief-function-based fault diagnosis in industrial settings?

The remainder of this thesis is structured as follows. Section 2 introduces the theoretical and practical background needed to understand the work, including fault diagnosis under uncertainty, Bayesian Networks [In what order should I present it?] , belief function theory, belief-function graphical models, and the related research gap. Section 3 defines the problem statement and scope of the thesis. Section 4 describes the electrical case study, the dataset, and the diagnostic task. Section 5 briefly presents the Bayesian Network baseline, which serves as the starting point for the belief-function implementation. Section 6 explains how the Bayesian Network model is transformed into a Dempster-Shafer belief-function model and how this implementation is carried out. Section 7 presents and analyses the diagnostic results. Section 8 discusses the practical applicability, scalability, software tooling, and technology readiness of the approach. Section 9 outlines possible directions for future work. Finally, Section 10 concludes the thesis by answering the research question.

2 Background and Related Work

2.1 Fault diagnosis under uncertainty

Fault diagnosis is a central task in the supervision of engineered systems because it links observations to explanations and actions. In industrial environments, the objective is not only to detect that abnormal behavior has occurred, but also to identify the most plausible fault causes early enough to support intervention, maintenance planning, or risk mitigation [20, 6]. The importance of this task increases with system complexity, tighter safety requirements, and stronger pressure to reduce downtime and maintenance cost.

The fault diagnosis literature is broad and heterogeneous. Classical reviews group diagnostic methods into quantitative model-based, qualitative model-based, and process-history-based approaches [20, 18, 19]. More recent surveys use the categories model-based, signal-based, knowledge-based, hybrid, and active fault diagnosis [4]. This classification is useful because it shows that no single diagnostic paradigm possesses all desirable industrial properties at once. In practice, real systems often require combinations of physical knowledge, data, rules, and uncertainty-aware reasoning.

The present work focuses on knowledge-based fault diagnosis under incomplete information. This focus is motivated by a recurring industrial situation: the structure of the causal or diagnostic model may be available, but the quantitative information needed to instantiate that model is only partially known. The present work focuses on knowledge-based fault diagnosis under incomplete information. This focus is motivated by a recurring industrial situation: the structure of the causal or diagnostic model may be available, but the quantitative information needed to instantiate that model is only partially known. Some probabilities may be unavailable, some fault-symptom relations may be supported only qualitatively, and some evidence sources may be incomplete, conflicting, or only partially reliable. Recent work on fusion-learning of Bayesian network models for fault diagnostics also illustrates that diagnostic knowledge may come from heterogeneous sources that need to be integrated

into a single reasoning model [1]. Under such conditions, the problem is not merely one of classification accuracy; it is also a problem of how uncertainty should be represented and propagated.

2.2 Bayesian probability and Bayesian networks

The Bayesian approach represents uncertainty by means of precise probabilities. Given a hypothesis H and evidence E , the posterior probability is obtained through Bayes' theorem:

$$P(H | E) = \frac{P(E | H)P(H)}{P(E)}. \quad (1)$$

This framework is attractive because it is mathematically coherent, decision-oriented, and well suited to combining prior knowledge with observed data [5].

A Bayesian network (BN) is a directed acyclic graph together with local conditional probability distributions. If $V_1, \dots, V_n$ are the network variables and Pa_i denotes the parents of node V_i , then the joint distribution factorizes as

$$P(v_1, \dots, v_n) = \prod_{i=1}^n P(v_i | pa_i). \quad (2)$$

This factorization follows from the local Markov condition according to which each variable is independent of its non-descendants given its parents [11, 2]. In fault diagnosis, this representation is especially appealing because it supports a modular encoding of causal relations between faults, internal states, and observable symptoms. Bayesian networks have also been used directly for fault detection and diagnosis in dynamic systems, where hidden system states must be inferred from noisy or imperfect observations [8].

Bayesian networks have several well-known strengths. They provide interpretable graphs, support reasoning with causal and diagnostic directions, and permit principled integration of expert knowledge and data. They are also widely used in modern industrial analytics because they bridge mechanism-based understanding and data-driven learning [5, 24, 10, 17]. In recent application reviews, diagnostic Bayesian networks are presented as promising tools for real-world fault detection and diagnosis because of their interpretability, robustness to uncertainty, scalability, and flexibility [10].

However, the Bayesian framework imposes an important requirement: before reasoning can begin, the model must be probabilistically complete. In practice, this means that priors and all conditional probability table (CPT) entries must be specified. This requirement becomes problematic when the Bayesian network structure is available but some parameters are unknown, weakly justified, or impossible to estimate reliably. Missing parent-state combinations may never occur in the data, and experts may be unwilling to provide precise probabilities for rare or poorly understood situations [14]. In such cases, standard Bayesian practice often relies on artificial completion rules such as equally likely priors, maximum-entropy completions, or sensitivity analysis. These strategies produce precise numbers, but they do not preserve ignorance explicitly.

This limitation is central to the present thesis. The motivation is not to replace Bayesian networks where precise and credible probabilities are available. Rather, the motivation is

to investigate what should be done when the graph structure is **useful**, but the precision demanded by Bayesian probability is not justified by the available knowledge.

2.3 Belief functions and Dempster–Shafer theory

The Dempster–Shafer (D–S) theory of belief functions, also referred to as **evidence theory**, offers a more flexible language for representing incomplete information [3, 13]. Let Θ denote a finite frame of discernment, i.e. the set of mutually exclusive and exhaustive hypotheses. A basic probability assignment (BPA), or mass function, is a mapping

$$m : 2^\Theta \rightarrow [0, 1] \quad (3)$$

such that

$$m(\emptyset) = 0, \quad \sum_{A \subseteq \Theta} m(A) = 1. \quad (4)$$

The subsets $A \subseteq \Theta$ such that $m(A) > 0$ are called focal elements. A key difference from probability theory is that mass **can be assigned to subsets**, not only to singleton hypotheses. This makes it possible to state that the evidence supports a set of hypotheses without forcing a precise distribution among them.

From the BPA, one defines the belief and plausibility functions:

$$Bel(A) = \sum_{B \subseteq A} m(B), \quad (5)$$

$$Pl(A) = \sum_{B \cap A \neq \emptyset} m(B) = 1 - Bel(\bar{A}). \quad (6)$$

For any proposition A , $Bel(A)$ is the amount of support committed to A , while $Pl(A)$ is the amount of support compatible with A . Hence, the interval

$$[Bel(A), Pl(A)] \quad (7)$$

can be interpreted as the range of probabilities consistent with the available evidence [13, 14]. In the Bayesian special case, where all focal elements are singletons, belief and plausibility collapse to a single additive probability distribution.

The ability to represent ignorance is the main conceptual advantage of belief functions for the present work. Complete ignorance is represented by the vacuous BPA

$$m(\Theta) = 1. \quad (8)$$

This representation has no genuine counterpart in standard Bayesian probability, where total mass must always be distributed over singleton hypotheses. A belief-function model can therefore distinguish between “lack of evidence for A ” and “evidence against A ,” whereas a precise probability model forces these cases into the same numerical space [16, 14].

When two distinct bodies of evidence are available, they can be combined by **Dempster’s rule**:

$$(m_1 \oplus m_2)(A) = \frac{\sum_{B \cap C = A} m_1(B)m_2(C)}{1 - \sum_{B \cap C = \emptyset} m_1(B)m_2(C)}, \quad A \neq \emptyset. \quad (9)$$

The denominator removes the mass assigned to conflicting evidence and renormalizes the remaining support [13, 7]. This rule is powerful but must be used carefully. It is only justified for distinct pieces of evidence; otherwise, the combination may double-count information. In addition, highly conflicting evidence can lead to controversial or counter-intuitive results, which explains why conflict management remains an active research topic in the contemporary D–S literature [23].

[Add the pigmistic probability and transformation equation???)

2.4 Belief-function graphical models

Although belief functions are often introduced in flat fusion settings, they can also be organized in graphical models. Foundational work by Shafer, Shenoy, and Mellouli showed that belief functions can be propagated through local computation in qualitative Markov trees, thereby avoiding some of the computational burden of repeated brute-force combination [12]. This line of work established that belief-function reasoning can be structured modularly, much as probabilistic reasoning is structured by Bayesian networks.

Recent work has made the directed graphical case more explicit. Jiroušek, Kratochvíl, and Shenoy define conditional belief functions so that they can play the role in directed belief-function models that CPTs play in Bayesian networks [7]. This is essential for the present thesis because it provides the mathematical language needed to discuss a transformation from Bayesian networks to Dempster–Shafer belief-function models.

A major result in this direction is given by Shenoy [14]. Consider a Bayesian network with variables $X_1, \dots, X_n$. For each node X_i with parents Pa_i , the Bayesian CPT $P(X_i | Pa_i)$ can be transformed into a conditional belief function by means of Smets’ conditional embedding. The embedded conditionals are then combined using Dempster’s rule. Shenoy shows that, if the original Bayesian network is complete, the resulting joint BPA is Bayesian and corresponds exactly to the BN joint distribution:

$$\bigoplus_{i=1}^n m_{X_i|Pa_i} \iff \prod_{i=1}^n P(X_i | Pa_i). \quad (10)$$

Therefore, a complete Bayesian network can be represented exactly as a directed D–S belief-function model.

This exact correspondence is important because it shows that moving from a BN to a D–S model does not necessarily imply changing the model semantics when full probabilistic information is available. The D–S model is then simply a more general host formalism. The gain appears when the Bayesian model is incomplete. If some priors or CPT rows are unknown, the corresponding parts can be left vacuous or partially specified in the belief-function representation. The outcome is not an arbitrary point estimate, but an interval-valued result reflecting what is known and what remains unknown [14].

A related modeling issue is the notion of *distinctness*. In probabilistic graphical models, conditional independence supports factorization. In D–S graphical models, distinctness plays an analogous role for justified combination. This is particularly relevant in diagnosis, because many evidence items are derived from common sensors, common preprocessing chains, or common expert judgments. If such dependencies are ignored, a belief-function model can

become overconfident for the wrong reasons [7, 15]. Any practical guideline for converting Bayesian networks into D–S models must therefore pay explicit attention to double counting and source distinctness.

2.5 State of the art and research gap

The current state of the art is unevenly distributed between Bayesian and D–S approaches. Bayesian networks are already supported by a mature software and application ecosystem. Recent industrial papers and reviews propose structured workflows for BN construction, integration of expert knowledge and data, and deployment in realistic diagnosis scenarios [17, 10, 24]. This maturity makes Bayesian networks a natural starting point for industrial users.

By contrast, the D–S fault diagnosis literature is rich but fragmented. One part of the literature is theoretical and concerns conditional belief functions, graphical models, entropy, distinctness, and local computation [7, 14]. Another part is application-driven and focuses on sensor fusion or classifier fusion for specific domains such as machinery, motors, bearings, aerospace, and unmanned aerial systems [9, 22]. A third part is devoted to improved combination rules and conflict management, which remains an active issue because classical Dempster normalization may behave poorly under strong conflict [23].

A particularly relevant comparative reference is the study of Verbert, Babuška, and De Schutter [21]. Their conclusion is nuanced and highly relevant for this thesis: when the available information is almost complete and non-conflicting, Bayesian reasoning is often preferable because it is computationally lighter, clearer, and easier to adapt. When the information is incomplete or conflicting, the D–S result may be more informative, because it preserves uncertainty instead of forcing precise probabilities.

Taken together, the literature suggests that there is no shortage of D–S applications to fault diagnosis. The real gap lies elsewhere. There is still limited guidance for practitioners who already have a Bayesian network and need to know: (i) when precise Bayesian probabilities are no longer justified; (ii) how to convert a BN into a D–S belief-function model without double counting information; (iii) how to interpret interval-valued outputs operationally; and (iv) under what conditions the additional modeling effort is worthwhile for industrial deployment. Addressing that gap is the motivation of the present thesis.

In this sense, the contribution of this work is not to claim that Dempster–Shafer theory is universally superior to Bayesian networks. Rather, the contribution is to formulate a principled and practical guideline for the subset of diagnosis problems in which a Bayesian network structure is valuable, but the available quantitative information is incomplete, partially reliable, or too weakly justified to support a fully precise probabilistic model.

3 Problem Statement and Research Scope

(Needed? Should be included into intro or after the BN building explanation to then lead into the DS approach??)

Bayesian networks require complete specification of conditional probabilities, which is often not feasible in real-world applications. This thesis investigates how belief function

theory can be used to model and solve fault diagnosis problems when probabilities are incomplete or uncertain.

[TO FILL/TO BE COMPLETED]

4 Case Study and Dataset

This section introduces the diagnostic case study used in the thesis. The purpose of the section is not to provide a full engineering specification of the system or a complete description of the Bayesian Network construction. Those details are documented separately in the project repository and, where necessary, summarized in the appendix. Instead, this section describes the system at the level needed to understand the belief-function implementation and the diagnostic experiments.

4.1 Electrical System Description

The case study is based on a modular electrical system whose overall function is to operate a load, represented by a lamp. The system consists of ten modules connected in series: one power supply module, eight control modules, and one load module. Each module consists of an electrical circuit enclosed in a plastic box, and the modules are connected using unmarked cables with fast-connection terminals. The cables are visually indistinguishable, which makes the diagnostic task more realistic because the technician cannot rely only on visual cable identification. The physical description of the system specifies that the modules are connected in series and that the circuit inside each module can be inspected through a panel [TO ADD GITHUB REPO REFERENCE].

The power supply module contains a 12V lithium battery and a green LED connected in parallel with the source. The LED turns on when the 12V source is supplied. Each of the eight control modules contains a switch and a red switch indicator LED. The indicator LED turns on when the 12V line of that module is energized and the module switch is on. The load module contains the main lamp, which is the final output of the system, and a smaller lamp indicator connected in parallel with it. A diode is also present in the load module to block current if the polarity is reversed, but this thesis does not model diode faults because they are outside the selected diagnostic scope.

The system is intentionally simple, but it captures several important properties of real diagnostic systems. First, faults can propagate through the chain: an upstream fault can cause all downstream indicators and the final lamp to turn off. Second, different faults can produce similar observations. For example, a detached switch, a broken cable, or a missing upstream voltage can all prevent downstream modules from receiving power. Third, some observations may be unavailable, such as when indicator LEDs are removed or when no multimeter is available. These properties make the system suitable for comparing precise Bayesian inference with belief-function reasoning under incomplete information.

4.2 Fault and Observable Variables

The diagnostic model distinguishes between hidden fault variables, intermediate voltage variables, and observable variables. The fault variables represent possible root causes. They include a power supply short, battery exhaustion, switch detachment faults for the eight control modules, cable faults between consecutive modules, a cable fault between the last control module and the load module, and a lamp fault. In total, the Bayesian Network specification defines 20 fault nodes: one power supply short node, one battery node, eight switch fault nodes, nine cable fault nodes, and one lamp fault node [TO ADD GITHUB REPO REFERENCE].

The voltage variables represent the propagation of electrical power through the system. The node V_0 represents the voltage at the power supply output. The nodes V_1 to V_8 represent the post-switch voltage after each control module. These variables are discretized into two states, 12V and 0V, because the diagnostic model is discrete.

The observable variables represent what a technician or diagnostic agent can observe. These include the power supply LED, the eight switch indicator LEDs, the main lamp, the lamp indicator, and optional multimeter readings. The multimeter observations are only available in scenarios where the diagnostic agent has a multimeter. When LEDs are physically removed or when no multimeter is available, these variables are treated as unobserved rather than being assigned a special third state.

4.3 Dataset Construction and Assumptions

The dataset and Bayesian Network structure were designed from the physical system rather than from the diagnostic scenarios. This means that the model is intended to represent the system generally, while the scenarios are used for validation and experimentation. The structural specification explicitly states that the Bayesian Network must work in an open-world setting without knowing in advance which scenario is occurring, and that the selected scenarios are used for validation rather than as the source of the model design [TO ADD GITHUB REPO REFERENCE].

Several simplifying assumptions were made to keep the system suitable for a comparative uncertainty-reasoning thesis. The model does not include diode faults, LED reliability faults, or internal wiring faults inside individual modules. Faults are represented at the level of the battery, power supply short, switch detachment, inter-module cables, cable to the load, and lamp failure. Most conditional probability tables are deterministic and follow the physical logic of voltage propagation: voltage reaches a downstream module only if the upstream voltage is present, the relevant cable is functional, and the switch is connected. The probabilistic part of the model is limited mainly to fault priors and the conditional relation between the power supply short and battery exhaustion.

The numerical priors used in the Bayesian Network should not be interpreted as true industrial reliability values. Due to time constraints, and because the construction of the Bayesian Network is not the main contribution of this thesis, these probabilities were elicited from a large language model acting as a simulated domain expert. The purpose of this elicitation was not to obtain exact reliability estimates, but to create a coherent and useful toy-domain model for diagnostic reasoning. The selected values were therefore intended to

be physically plausible, internally consistent, and useful for producing non-trivial diagnostic ambiguity. They were selected to create a coherent and realistic toy-domain reliability model, while preserving enough ambiguity to compare different uncertainty-reasoning frameworks. The prior elicitation document explicitly states that the objective is not to produce true reliability engineering values, but to create a plausible educational model that behaves sensibly under diagnosis and contains enough uncertainty for comparing Bayesian Networks, belief functions, ranking theory, and related approaches [TO ADD GITHUB REPO REFERENCE].

4.4 Diagnostic Task

The diagnostic task is to infer the most plausible underlying fault or set of faults from a set of observations. Observations may include LED states, lamp states, lamp indicator states, and multimeter measurements when available. Some scenarios provide relatively rich evidence, while others intentionally remove diagnostic information, for example by removing all control-module LEDs or by making the multimeter unavailable.

The selected diagnostic scenarios include battery exhaustion, switch detachment in different control modules, missing control-module LEDs, absence of multimeter measurements, multi-fault cases, and a power supply short that also discharges the battery. These scenarios are useful because they create different levels of diagnostic ambiguity. In simple cases, the available evidence may strongly indicate one fault. In more ambiguous cases, the model should not overcommit to a single explanation. This is especially important for the belief-function implementation, because one of its expected advantages is the ability to preserve and communicate uncertainty rather than forcing a precise posterior probability.

The Bayesian Network baseline is therefore used as a structured starting point for the belief-function implementation. The detailed construction of the Bayesian Network, including the full directed acyclic graph, conditional probability tables, priors, validation checks, and scenario implementation, is documented in the project repository.

4.5 Bayesian Network Baseline

[TO FILL/TO BE COMPLETED]

The Bayesian Network baseline was developed collaboratively as part of the group project. Since the main contribution of this thesis is the belief-function implementation and analysis, the BN construction is summarized only briefly here. The full model specification and implementation can be found in the project repository: [TO FILL/TO BE COMPLETED: GitHub directory citation/link].

[TO FILL/TO BE COMPLETED]

5 Belief-Function Implementation

[TO FILL/TO BE COMPLETED]

5.1 From Bayesian Network to Belief-Function Model

[TO FILL/TO BE COMPLETED]

5.2 Mass Functions, Evidence, and Inference

[TO FILL/TO BE COMPLETED]

5.3 Implementation Using Belief Function Machine

Here I should have the TRL and the state of the art, explain which software I'm using and why [TO FILL/TO BE COMPLETED]

5.4 Worked Example

[TO FILL/TO BE COMPLETED]

6 Results and Analysis

[TO FILL/TO BE COMPLETED]

6.1 Experimental Setup

[TO FILL/TO BE COMPLETED]

6.2 Diagnostic Results

[TO FILL/TO BE COMPLETED]

6.3 Comparison with the Bayesian Network Baseline

[TO FILL/TO BE COMPLETED]

6.4 Interpretation of the Results

[TO FILL/TO BE COMPLETED]

7 Applicability and Scalability

[TO FILL/TO BE COMPLETED]

7.1 Practical Usefulness and Limitations

[TO FILL/TO BE COMPLETED]

7.2 Scalability and Bottlenecks

[TO FILL/TO BE COMPLETED]

7.3 Software and Tooling

[TO FILL/TO BE COMPLETED]

8 Future Work???

[TO FILL/TO BE COMPLETED]

9 Conclusion

[TO FILL/TO BE COMPLETED]

A Additional Dataset Details

[TO FILL/TO BE COMPLETED]

B Additional Bayesian Network Details

[TO FILL/TO BE COMPLETED]

C Additional Belief-Function Model Details

[TO FILL/TO BE COMPLETED]

D Additional Results

[TO FILL/TO BE COMPLETED]

References

- [1] T. T. Ademujimi and V. Prabhu. Fusion-learning of bayesian network models for fault diagnostics. *Sensors*, 2021.
- [2] A. P. Dawid. Conditional independence in statistical theory. *Journal of the Royal Statistical Society B*, 1979.
- [3] A. P. Dempster. Upper and lower probabilities induced by a multivalued mapping. *Annals of Mathematical Statistics*, 1967.

- [4] Z. Gao, C. Cecati, and S. Ding. A survey of fault diagnosis and fault-tolerant techniques. *IEEE Transactions on Industrial Electronics*, 2015.
- [5] David Heckerman. A tutorial on learning with bayesian networks. *Microsoft Research*, 1995.
- [6] Rolf Isermann. *Fault-Diagnosis Systems*. Springer, 2006.
- [7] R. Jirousek, V. Kratochvil, and P. P. Shenoy. On conditional belief functions in directed graphical models. *International Journal of Approximate Reasoning*, 2023.
- [8] Uri Lerner, Ronald Parr, Daphne Koller, and Gautam Biswas. Bayesian fault detection and diagnosis in dynamic systems. In *Proceedings of the Seventeenth National Conference on Artificial Intelligence*, 2000.
- [9] N. et al. Liu. Fault diagnosis of unmanned aerial systems using dempster-shafer theory. *Machines*, 2024.
- [10] C. Lu, Z. Wang, M. Mosteiro-Romero, and L. Itard. Diagnostic bayesian network in building energy systems. *Energy and Buildings*, 2025.
- [11] Judea Pearl. *Probabilistic Reasoning in Intelligent Systems*. Morgan Kaufmann, 1988.
- [12] G. Shafer, P. P. Shenoy, and K. Mellouli. Propagating belief functions in qualitative markov trees. *International Journal of Approximate Reasoning*, 1987.
- [13] Glenn Shafer. *A Mathematical Theory of Evidence*. Princeton University Press, 1976.
- [14] Prakash P. Shenoy. Making inferences in incomplete bayesian networks: A dempster-shafer belief function approach. *International Journal of Approximate Reasoning*, 2023.
- [15] Prakash P. Shenoy. On distinctness of belief functions. *Proceedings*, 2023.
- [16] P. Smets and R. Kennes. The transferable belief model. *Artificial Intelligence*, 1994.
- [17] M. Soleimani, S. Shahbeigi, and M. Nasr Esfahani. A bayesian network development methodology for fault analysis. *Expert Systems with Applications*, 2024.
- [18] V. Venkatasubramanian, R. Rengaswamy, and S. Kavuri. A review of process fault detection and diagnosis: Part ii. *Computers and Chemical Engineering*, 2003.
- [19] V. Venkatasubramanian, R. Rengaswamy, and S. Kavuri. A review of process fault detection and diagnosis: Part iii. *Computers and Chemical Engineering*, 2003.
- [20] V. Venkatasubramanian, R. Rengaswamy, K. Yin, and S. Kavuri. A review of process fault detection and diagnosis: Part i. *Computers and Chemical Engineering*, 2003.
- [21] K. Verbert, R. Babuska, and B. De Schutter. Bayesian and dempster-shafer reasoning in fault diagnosis. *IFAC-PapersOnLine*, 2017.

- [22] B.-S. Yang and K. J. Kim. Application of dempster-shafer theory in fault diagnosis. *Mechanical Systems and Signal Processing*, 2004.
- [23] H. et al. Zhao. A survey on conflict management in dempster-shafer theory. *Information Fusion*, 2022.
- [24] J. Zheng, Y. Zhuo, X. Jiang, L. Zeng, and Z. Ge. Advances in bayesian networks for industrial process analytics. *Expert Systems with Applications*, 2025.